

auxi-mpp: Material Physical Property Models for Pyrometallurgy in Python

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

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Editor: [✉](#)

Submitted: 20 May 2026

Published: unpublished

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Summary

In modern pyrometallurgy, computational models are key for the development, design, and operation of efficient and sustainable low-CO₂ smelting furnaces. However, the models that describe the material physical properties that govern the behaviour of these reactors are scattered across the literature and difficult to re-implement and maintain. To address this, auxi-mpp implements property models for key materials in high-temperature processing, including molten slags and alloys, as well as process gas. It consolidates a set of property models (e.g., density, viscosity, electrical conductivity) and documents their theoretical foundations. The software is designed for straightforward integration with multiphysics solvers, and auxi-mpp enables temperature-, pressure-, and composition-dependent property updates in pyrometallurgical process simulations.

Statement of need

With pyrometallurgical industries being the cornerstone of many countries around the globe, it is of global economic interest for these industries to thrive. However, they are currently facing great challenges which include stricter environmental regulations, such as the EU's target of a 30% reduction in CO₂ emissions by 2030, and the rising global demand for high-grade steel with increased reliance on lower-quality ores due to the depletion of high-grade resources. Due to these pressures, the traditional timeline for developing and qualifying new metallurgical processes, which often spans decades, is no longer adequate. This calls for accelerating innovation, requiring an advancement in the tools used, and signalling a need to describe these processes more fundamentally. To accelerate development cycles and reduce scale-up risks for emerging technologies, computational modelling, which includes coupled multiphysics simulations (CFD/FEM/DEM) and digital twins, became critical to rely on.

However, the capability of any high-fidelity computational model is ultimately constrained by the quality of its input parameters, of which material physical properties are key. In high-temperature reactors, which often operate above 1500 °C, process behaviour is governed by a coupled interplay of physical and chemical properties that evolve dynamically with temperature and composition. These properties, specifically viscosity, density, thermal and electrical conductivity, are the hidden foundation determining process viability as they are linked directly to key physical phenomena such as chemical reactions, bath mixing, phase separation, and refractory erosion. Despite this critical reliance on material properties, the industry is facing a substantial knowledge gap, and the material property models that do exist are scattered across the literature and are only available as equations on paper.

To address this, auxi-mpp was developed, taking the first logical step in addressing the need for material properties by consolidating models available in the literature into a Python package

that can be easily accessed and used. The significance of these implemented models is that they reduce uncertainty in process and multiphysics simulations, making them more realistic and valuable. By linking process conditions and material properties directly, `auxi-mpp` strengthens our ability to de-risk the next generation smelting technologies with a strong and credible basis of computational models. `auxi-mpp` can be used to do direct calculations in Python, or its models can be built into computational models to provide dynamic material property updates (see Figure 1).

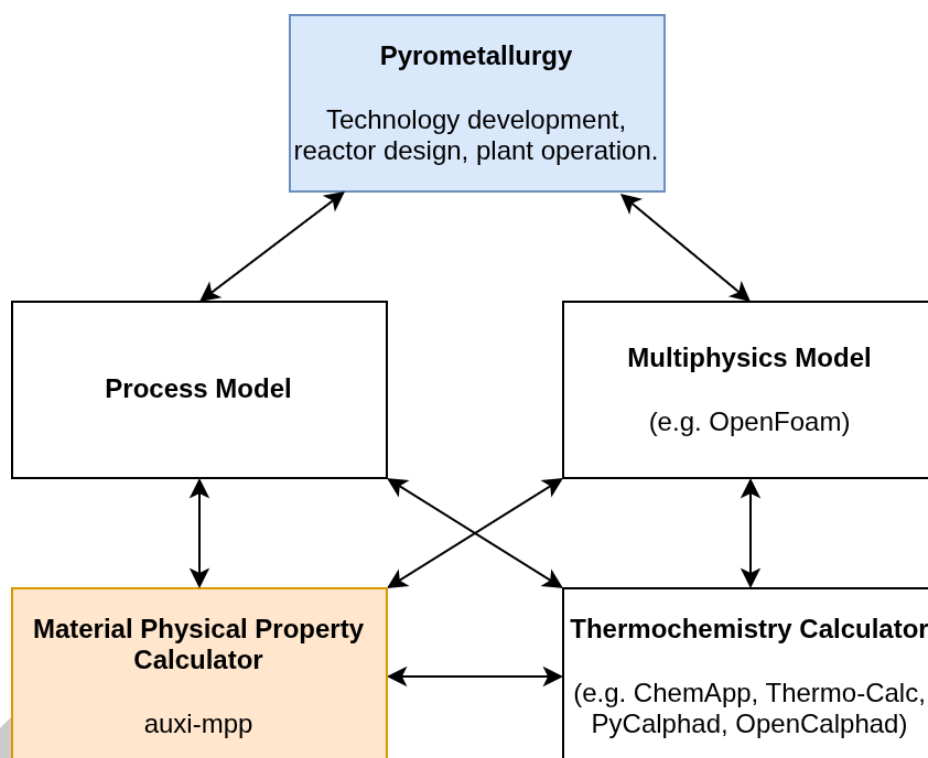


Figure 1: The context in which `auxi-mpp` is intended to be used.

State of the field

Other software that can estimate material physical properties do exist, but are limited and hard to find. The best-known commercial software packages in pyrometallurgy that have some material physical property descriptions are FactSage (Bale et al., 2016) and Thermo-Calc (Andersson et al., 2002), which focus mainly on thermochemical calculations, but allow some material physical properties estimations. Open-source CALPHAD tools such as PyCalphad and OpenCalphad likewise target phase equilibria rather than transport and thermophysical properties.

Software design

Software architecture

`auxi-mpp` is structured as a modular Python package designed to promote extensibility and type safety. It relies on `auxi-core` and `auxi-chemistry`, other packages in the `auxi family`, for base data structures, input validation, and stoichiometry calculations. The core architecture follows a hierarchical Object-Oriented Programming (OOP) paradigm, with final, usable physical property models being derived from a set of abstract base classes.

The inheritance structure, shown in Figure 2, promotes code re-use and consistency, limits code duplication, and enforces constraints such as units and material states. The hierarchy proceeds as follows:

1. `MaterialPropertyModel` is the root abstract base class from which all others are derived. It defines the standard interface for all models, which includes the property name, units, LaTeX display symbols, and references.
2. The next layer distinguishes between material states and types, and includes abstract base classes for gases and condensed materials, which are further specialised into abstract classes for liquid alloys and slags.
3. The last abstract base class layer specifies the property being described, its units, and display symbol (e.g., $\text{m}^3 \text{mol}^{-1}$ and $\bar{V}$ for molar volume).
4. The final layer consists of concrete, usable classes that implement the algorithms found in literature (e.g., `ThibodeauBinary`). To improve computational performance, separate implementations are often provided for unary, binary, and multi-component systems.

Legend

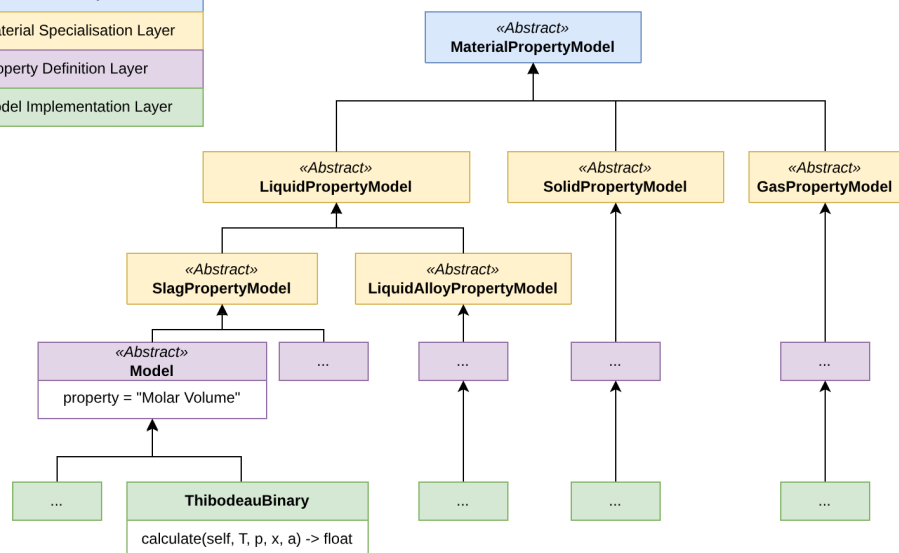
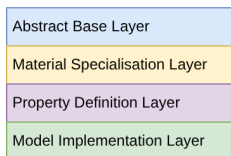


Figure 2: auxi-mpp class inheritance hierarchy.

To ensure robustness, the architecture utilises Python generics to enforce type hinting on thermodynamic variables, specifically temperature (T), pressure (p), and composition (x).

A critical architectural decision in auxi-mpp is the stateless nature of model instances meaning that computation is separated from configuration. The standard flow of usage is as follows:

1. **Initialisation:** When a model class is instantiated, it loads all necessary model-specific coefficients and datasets into memory. This is a one-time cost.
2. **Calculation:** The `calculate` method is pure and stateless. It accepts the temperature, pressure and composition as arguments and returns the property value without modifying instance attributes.

This design allows a single model instance to be reused safely across many iterative calculations (common in process and multiphysics simulation loops) without the overhead of re-initialising data or the risk of state contamination between steps.

This flexible framework is supported by a high-performance backend leveraging Polars, NumPy, and SciPy, ensuring efficient batch processing and numerical accuracy across wide temperature

and composition ranges.

Software functionalities

auxi-mpp currently provides an interface to estimate thermophysical and transport properties of high-temperature materials, with an emphasis on slag. All models estimate the physical properties from temperature and composition inputs. Table 1 summarises the property models available for slag consisting of at least Al_2O_3 , CaO , Fe_2O_3 , FeO , MgO , SiO_2 . All quantities in auxi-mpp strictly use SI units.

Table 1: Slag material property models currently available in auxi-mpp. All slag systems are combinations of Al_2O_3 , CaO , Fe_2O_3 , FeO , MgO and SiO_2 .

Physical Property	Description	Units	Reference
Density (Molar Volume)	A structure-based model that uses equilibrium calculation results from a thermochemistry calculator such as ChemApp and data from the FactSage FToxid database.	kg m^{-3} ($\text{m}^3 \text{mol}^{-1}$)	(Thibodeau et al., 2016)
Dynamic Viscosity	Grundy and Kim Model: A structure-based model that also uses MQM results based on the FactSage FToxid database.	Pa s	Kim et al. (2021)
	Zhang Model: A structure-based model independent of MQM results.	Pa s	(Zhang et al., 2014)
Electrical Conductivity	Thibodeau Model: An implementation of the Nernst–Einstein relationship to estimate ionic conductivity from molar volume and ionic diffusivity.	S m^{-1}	(Thibodeau, 2016)
	Hundermark Model: An empirical model obtained by applying regression techniques based on experimental data.	S m^{-1}	(Hundermark, 2003)

Table with 4 columns: Physical Property, Description, Units, Reference. Rows include Zhang Model (electrical conductivity) and Ionic Diffusivity (Arrhenius equation).

Density models use a simple algebraic conversion of the results from the molar volume models. auxi-mpp also provides models for solids, liquid alloys, gases and multiphase materials. For solids there are models available for density and thermal conductivity. For liquid alloy there are models for density (molar volume), viscosity, and electrical and thermal conductivity. For gases, models are implemented for density, viscosity, thermal conductivity, and total emissivity. There are two sets of documentation that accompany auxi-mpp: the user documentation and the Theory Manual. The user documentation contains all that is needed to get started and use the models in auxi-mpp and is available directly from the GitLab page. The Theory Manual, available from the user documentation webpage, explains how the models estimate the physical properties from the temperature and composition inputs, and documents the performance of the models, explicitly commenting on possible limitations. All models are validated against experimental data and, where possible, against the performance as documented by the original authors. These validation figures can also be found in the Theory Manual.

Research impact statement

Decarbonisation presents the challenge of phasing out highly successful, established technologies and replacing them with greener alternatives. A striking example in pyrometallurgy is the work of the Groeien met Groen Staal (GGS) consortium in The Netherlands. To meet EU regulations, GGS aims to replace the blast furnace, a technology refined over more than 2,000 years, with sustainable alternatives (Groeien met Groen Staal, 2024). To accelerate this research and development, GGS has invested heavily in computational modelling. However, the efficacy of these numerical models depends on the availability of reliable data; accurate thermochemical and thermophysical properties of molten silicate slags are essential for optimising metallurgical processes (Matsushita et al., 2005). Furthermore, a deep understanding of slag structure is fundamental to deciphering slag properties and advancing the science of iron- and steelmaking (Seetharaman et al., 2013). Recognising that robust material descriptions are the backbone of high-fidelity modelling, GGS funded the development of auxi-mpp, which is now a valuable tool for engineers developing multiphysics models of high-temperature processes. It streamlines the modelling workflow by providing composition- and temperature-dependent properties that can be updated dynamically during simulations. By eliminating the need for engineers to manually search literature and implement property models, auxi-mpp significantly reduces the time required to build sophisticated and more realistic simulations. It also allows engineers and scientists to critically evaluate model outputs based on the provided theoretical documentation. In a recent conference paper, auxi-mpp was highlighted as a framework to support consistent material property evaluation (Zhu et al., 2026). As of 3 August 2026, auxi-mpp has reached 11,710 downloads since its release on 27 May 2025. By providing transparent and validated material descriptions, the package frees engineers to focus on innovation in the pyrometallurgical

industry rather than manual re-implementation of complex models. This makes `auxi-mpp` a valuable and convenient toolkit for modern pyrometallurgical innovation, supporting the industry's shift towards more efficient and sustainable operations.

AI usage disclosure

During development, Gemini 3 was used to write and refactor portions of the `auxi-mpp` source code, tests, and user documentation. Nevertheless, the final code was reviewed and revised by the human authors, and all design choices were made by them as well.

During the preparation of this paper, the authors used Gemini 3 to provide an initial description of the software architecture. After this, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Conflict of Interests

The authors have no conflict of interests to disclose.

Acknowledgements

We acknowledge the Groeien met Groen Staal (GGS) consortium and the National Growth Fund (NGF) of the Netherlands for providing funding. We thank our previous colleague Stefan van der Merwe for setting up the digital infrastructure.

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